

Lesson 5

Probability: Theory vs Experiment

Exploring the Exciting World of Chance!

Grade 4 · Academic Year 2025–26

Learning Objectives

1	Define probability and use it to describe likelihood of events.
2	Calculate theoretical probability using the formula $P = \text{favourable} / \text{total}$.
3	Record and calculate experimental probability through hands-on trials.
4	Compare theoretical and experimental probability and explain differences.
5	Apply probability thinking to real-world situations.

What Is Probability?

Probability is the mathematics of chance. It tells us how likely an event is to happen on a scale from 0 (impossible) to 1 (certain). Understanding probability helps us make better predictions about uncertain events in everyday life — from weather forecasts to board games!

The Probability Scale

Value	Meaning	Example
0	Impossible — will never happen	Rolling a 7 on a standard die
0.25	Unlikely — might happen rarely	Picking a red marble from 1 red + 3 blue
0.5	Even chance — 50/50	Flipping heads on a fair coin
0.75	Likely — will probably happen	Picking blue from 3 blue + 1 red
1	Certain — will always happen	Rolling a number < 7 on a die

Theoretical Probability

Theoretical probability is what we calculate using mathematics BEFORE running any experiment. It tells us what SHOULD happen in a perfectly fair situation.

Formula: $P(\text{event}) = \text{Number of favourable outcomes} \div \text{Total number of equally likely outcomes}$

Theoretical Probability Examples

Experiment	Event	Favourable	Total	P (fraction)	P (%)
Flip a coin	Heads	1	2	1/2	50%
Roll a die	Roll a 4	1	6	1/6	≈17%
Roll a die	Roll an even number	3	6	3/6 = 1/2	50%
Bag: 2R + 3B marbles	Pick red	2	5	2/5	40%
Spin 4-colour wheel	Land on green	1	4	1/4	25%

—■ Activity: Calculate It!

A bag has 3 yellow, 2 green, and 5 red balls. Calculate the theoretical probability of:

(a) Picking a yellow ball = _____ (b) Picking a green ball = _____ (c) Picking red = _____

Check: Do all three probabilities add up to 1 (or 100%)? _____

Experimental Probability

Experimental probability is what ACTUALLY happens when we run a real experiment. It is calculated from recorded results, not from maths formulas. Real experiments can give surprising results — especially when the number of trials is small!

Formula: $P(\text{event}) = \text{Number of times event occurred} \div \text{Total number of trials}$

Sample Coin-Flip Experiment (20 flips)

Trial Group	Heads	Tails	Exp. P(Heads)	Theoretical P(Heads)
Flips 1–5	3	2	3/5 = 60%	50%
Flips 6–10	2	3	2/5 = 40%	50%

Flips 11–20	6	4	$6/10 = 60\%$	50%
ALL 20 flips	11	9	$11/20 = 55\%$	50%

—■ Activity: Coin Flip Experiment

You will need: a coin and a tally sheet.

Flip the coin 30 times. Record each result (H or T) using tally marks.

After all 30 flips, count: Heads = ____ Tails = ____

Calculate: Experimental P(Heads) = ____/30 = ____%

Compare with theoretical P(Heads) = 50%. Are they the same? _____

What do you think would happen if you flipped 300 times instead of 30?

Comparing Theoretical vs Experimental

Key differences at a glance:

THEORETICAL PROBABILITY	EXPERIMENTAL PROBABILITY
Calculated using maths formulas	Calculated from real trial results
Based on perfect, fair conditions	Based on actual, imperfect reality
Always the same for the same experiment	Can vary each time you run the experiment
Tells us what SHOULD happen	Tells us what DID happen
Useful for predictions before an experiment	Useful for checking if conditions are really fair

The Law of Large Numbers: The more trials you run, the closer your experimental probability gets to the theoretical probability!

This is why casinos, insurance companies, and weather forecasters are so accurate — they use millions of data points. With only a few trials, chance plays a big role. With thousands of trials, results settle very close to the theoretical prediction.

Probability in Real Life

Probability is everywhere around us:

- Weather Forecasts — '70% chance of rain' means in 100 similar weather situations, it rained 70 times.
- Board Games — dice games, spinners, and card games all rely on probability.

- Sports — coaches use probability to decide which player should take a penalty kick.
- Medicine — doctors use probability to understand how likely a treatment is to work.
- Insurance — companies calculate probability of accidents to set fair prices.
- Shopping — sale discounts use probability of how likely you are to buy more.

—■ Activity: Spinner Investigation

Design a spinner with 4 equal sections: Red, Blue, Green, Yellow.

Theoretical probability of landing on each colour = $1/4 = 25\%$.

If you spin 40 times, how many times do you EXPECT each colour? _____ each.

Spin 40 times and record results. Did you get exactly 10 each? _____

Why might results differ from the theoretical prediction?

Quick Check ✓

1. A die is rolled once. What is the theoretical probability of rolling a number greater than 4? Show your working.
2. You flip a coin 50 times and get 28 heads. What is the experimental probability of heads? Is this close to theoretical?
3. Explain in your own words why experimental probability might differ from theoretical probability.
4. A bag has 4 red, 3 blue, and 3 green marbles. What is the probability of NOT picking red?
5. Why does experimental probability get closer to theoretical probability as you do more trials?

Lesson Summary

- ★ Probability measures likelihood on a scale from 0 (impossible) to 1 (certain).
- ★ Theoretical $P = \text{Favourable outcomes} \div \text{Total outcomes}$ (calculated before the experiment).
- ★ Experimental $P = \text{Times event occurred} \div \text{Total trials}$ (measured from real results).
- ★ The two types rarely match perfectly — especially with few trials.
- ★ The more trials you run, the closer experimental probability gets to theoretical (Law of Large Numbers).